

TalentSphere AI AI Powered Human Resource Management System

Hackathon Project Blueprint

June 2026

Contents

1	Project Overview	3
2	Objectives	3
3	Technology Stack	3
3.1	Frontend	3
3.2	Backend	3
3.3	AI Service	3
3.4	Database	3
3.5	Deployment	4
4	System Architecture	4
5	User Roles	4
5.1	Management Admin	4
5.2	Senior Manager	4
5.3	HR Recruiter	4
5.4	Employee	5
6	Core HRMS Features	5
6.1	Employee Management	5
6.2	Attendance Management	5
6.3	Leave Management	5
6.4	Payroll Management	5
6.5	Performance Management	5
7	Onboarding Management	6
7.1	Candidate Conversion	6
7.2	Document Verification	6
7.3	Onboarding Progress Tracker	6
7.4	AI Onboarding Assistant	6
7.5	AI Generated Welcome Kit	7
8	AI Features	7
8.1	AI Feature 1: Resume Screening	7
8.2	AI Feature 2: Candidate Ranking	7
8.3	AI Feature 3: Video Interview Analysis	8
8.4	AI Feature 4: HR Chatbot	8
8.5	AI Feature 5: Sentiment Analysis	8

8.6 AI Feature 6: Attrition Risk Prediction	8
9 Analytics Dashboard	9
10 Database Tables	9
11 Development Roadmap	9
11.1 Day 1	9
11.2 Day 2	9
11.3 Day 3	10
11.4 Day 4	10
11.5 Day 5	10
11.6 Day 6	10
11.7 Day 7	10
12 Hackathon Demo Flow	10
13 Future Enhancements	11

1 Project Overview

TalentSphere AI is a next-generation AI-powered Human Resource Management System (HRMS) designed to automate recruitment, employee management, onboarding, payroll, performance evaluation, and workforce analytics.

The system combines traditional HRMS functionalities with Artificial Intelligence to reduce manual effort and improve decision making.

2 Objectives

- Automate HR operations
- Implement AI-driven recruitment
- Provide intelligent employee insights
- Support multiple user roles
- Deliver scalable architecture
- Provide responsive web experience

3 Technology Stack

3.1 Frontend

- Next.js
- React.js
- Tailwind CSS
- ShadCN UI

3.2 Backend

- Node.js
- Express.js

3.3 AI Service

- Python FastAPI
- Gemini API
- HuggingFace Transformers
- Whisper Speech-to-Text

3.4 Database

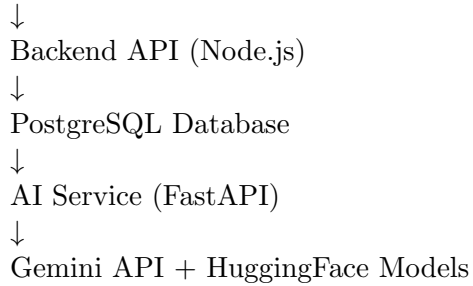
- PostgreSQL

3.5 Deployment

- Vercel
- Render

4 System Architecture

Frontend (Next.js)



5 User Roles

5.1 Management Admin

- Manage employees
- Manage recruiters
- Company analytics dashboard
- Payroll monitoring
- Attrition prediction reports

5.2 Senior Manager

- Team performance monitoring
- Team attendance
- Team leave approvals

5.3 HR Recruiter

- Create job openings
- Resume screening
- Candidate ranking
- Interview management
- Onboarding management

5.4 Employee

- View profile
- Attendance records
- Apply leave
- Download payslips
- Chat with HR Assistant

6 Core HRMS Features

6.1 Employee Management

- Add Employee
- Edit Employee
- Delete Employee
- Department Assignment
- Skill Tracking

6.2 Attendance Management

- Daily Check-In
- Daily Check-Out
- Attendance Reports

6.3 Leave Management

- Leave Application
- Leave Approval
- Leave History

6.4 Payroll Management

- Salary Calculation
- Tax Deduction
- PDF Payslip Generation

6.5 Performance Management

- KPI Tracking
- Employee Reviews
- Performance Reports

7 Onboarding Management

The onboarding module automates the process of converting a selected candidate into an employee.

7.1 Candidate Conversion

- Convert shortlisted candidate to employee
- Generate Employee ID
- Assign Department
- Assign Manager
- Assign Role

7.2 Document Verification

- Aadhaar Verification
- PAN Verification
- Educational Certificates
- Experience Documents
- Offer Letter Acceptance

7.3 Onboarding Progress Tracker

- Pending Tasks
- Completed Tasks
- HR Approval Status
- Joining Status

7.4 AI Onboarding Assistant

The AI assistant helps newly joined employees understand company policies and complete onboarding tasks.

Capabilities:

- Explain company policies
- Explain leave policies
- Explain payroll structure
- Guide document submission
- Answer onboarding questions

Workflow:

Candidate Selected

↓

Offer Letter Generated

↓

Document Submission

↓

AI Assisted Onboarding

↓

Employee Account Created

↓

Employee Activated

7.5 AI Generated Welcome Kit

Using Gemini API:

- Personalized Welcome Message
- Team Introduction
- Department Overview
- First Week Checklist

8 AI Features

8.1 AI Feature 1: Resume Screening

Goal: Automatically evaluate resumes without human intervention.

Workflow:

Resume Upload

↓

Text Extraction

↓

Skill Detection

↓

Job Matching

↓

Candidate Score

Outputs:

- Match Percentage
- Missing Skills
- Recommendation

8.2 AI Feature 2: Candidate Ranking

Automatically rank all candidates.

Output:

1. Candidate A - 91

2. Candidate B - 86
3. Candidate C - 74

8.3 AI Feature 3: Video Interview Analysis

Workflow:

Candidate Records Video

↓

Whisper Transcription

↓

Gemini Analysis

↓

Score Generation

Evaluation Parameters:

- Communication
- Technical Knowledge
- Confidence
- Sentiment

8.4 AI Feature 4: HR Chatbot

Capabilities:

- Leave Queries
- Payroll Queries
- HR Policies
- Employee Information

8.5 AI Feature 5: Sentiment Analysis

Analyze employee feedback.

Output:

- Positive
- Neutral
- Negative

8.6 AI Feature 6: Attrition Risk Prediction

Predict employee resignation probability.

Output:

- Low Risk
- Medium Risk
- High Risk

9 Analytics Dashboard

- Employee Count
- Hiring Trends
- Attendance Trends
- Payroll Statistics
- Attrition Reports
- Recruitment Funnel

10 Database Tables

1. Users
2. Employees
3. Attendance
4. Leave Requests
5. Payroll
6. Jobs
7. Resumes
8. Interviews
9. Performance Reviews
10. Feedback

11 Development Roadmap

11.1 Day 1

- Project setup
- Database setup
- Authentication
- Role management

11.2 Day 2

- Employee management
- Attendance system
- Leave management

11.3 Day 3

- Payroll module
- Dashboard UI

11.4 Day 4

- Resume screening AI
- Candidate ranking AI

11.5 Day 5

- Interview analysis
- HR chatbot

11.6 Day 6

- Sentiment analysis
- Attrition prediction

11.7 Day 7

- Testing
- Deployment
- Documentation
- Demo recording

12 Hackathon Demo Flow

1. Login as HR
2. Upload 20 resumes
3. Show AI ranking
4. Launch AI interview analysis
5. Show chatbot interaction
6. Switch to Admin dashboard
7. Show company analytics
8. Show attrition prediction

13 Future Enhancements

- Face Recognition Attendance
- Voice Controlled HR Assistant
- Skill Gap Analysis
- Personalized Learning Recommendations